

Course Code: CSE2051	Course Title: Information Retrieval Type of Course: Theory Only Course	L- T- P- C	3 -0	0	3
Version No.	1				
Course Pre-requisites	Basic Knowledge in Data Structures and algorithms and probability and statistics, background in machine learning				
Anti-requisites	NIL				
Course Description	The course studies the theory, design and implementation of Text- based information systems. The Information Retrieval core concepts of the course include statistical characteristics of text, representation of information needs and documents. Topics Include Several important retrieval models (Basic IR Models, Boolean Model, TF-IDF (Term Frequency/Inverse Document Frequency) Weighting, Vector Model, Probabilistic Model, Latent Semantic Indexing Model, Neural Network Model). Retrieval Evaluation, Retrieval Metrics, Text Classification and Clustering algorithms, Web Retrieval and Crawling. Recommender Systems: Basics of Content-based Recommender Systems, Content-based Filtering, Collaborative Filtering, Matrix factorization models and neighborhood models.				
Course Objective	The objective of the course is to familiarize the learners with the concepts Information Retrieval and attain Skill Development through Participative Learning techniques.				
Course Out Comes	On successful completion of the course the students shall be able to: CO1: Define basic concepts of information Retrieval. [Knowledge] CO2: Evaluate the effectiveness and efficiency of different information retrieval methods. [Application] CO3: Explain different indexing methodology requirements and the concept of web retrieval and crawling. [Comprehension] CO4: Classify different recommender system and its aspect. [Comprehension]				
Course Content:					
Module 1	Introduction to Information Retrieval	Assignment	Data collection	7	Sessions
Information Retrieval – Early Developments – The IR Problem – The Users Task – Information versus Data Retrieval – The IR System – The Software Architecture of the IR System – The Retrieval and Ranking Processes					
Module 2	Modeling and Retrieval Evaluation	Assignment	Problem solving	10	Sessions
Basic IR Models – Boolean Model – TF-IDF (Term Frequency/Inverse Document Frequency) Weighting – Vector Model – Probabilistic Model – Latent Semantic Indexing Model – Neural Network Model – Retrieval Evaluation – Retrieval Metrics – Precision and Recall – Reference Collection – User-based Evaluation – Relevance Feedback and Query Expansion – Explicit Relevance Feedback.					

Module 3	Indexing & Web-Retrieval	Term paper/Assignment	Data analysis	8 Sessions
Indexing and Searching – Inverted Indexes – Sequential Searching – Multi-dimensional Indexing. The Web – Search Engine Architectures – Cluster based Architecture - Search Engine Ranking – Link based Ranking – Simple Ranking Functions, Evaluations — Search Engine Ranking – Applications of a Web Crawler.				
Module 4	Recommender System	Term paper/Assignment	Problem solving	8 Sessions
Recommender Systems Functions – Data and Knowledge Sources – Recommendation Techniques – Basics of Content-based Recommender Systems – High Level Architecture – Advantages and Drawbacks of Content-based Filtering – Collaborative Filtering – Matrix factorization models.				
Targeted Application & Tools that can be used:				
Information Retrieval System, Collaborative Filtering System, Feedback System, Evaluation Metrics				
Assignment:				
Group assignment, Quiz				
Text Book				
T1 Ricardo Baeza-Yates and Berthier Ribeiro-Neto, —" Modern Information Retrieval: The Concepts and Technology behind Search", Third Edition, ACM Press Books, 2018. Link: https://people.ischool.berkeley.edu/~hearsst/irbook/				
T2 Ricci, F, Rokach, L. Shapira, B.Kantor, —"Recommender Systems Handbook", Fourth Edition, 2018.				
References				
R1 Stefan Buettcher, Charles L. A. Clarke and Gordon V. Cormack, —"Information Retrieval: Implementing and Evaluating Search Engines", The MIT Press, 2017.				
R2 Jian-Yun Nie Morgan & Claypool –" Cross-Language Information Retrieval", Publisher series 2011.				
R3 Stefan M. R�ger Morgan & Claypool – "Multimedia Information Retrieval", Publisher series 2014.				
R4 B. Liu, Springer, - "Web Data Mining: Exploring Hyperlinks, Contents, and Usage Data", Second Edition, 2013.				
R5 C. Manning, P. Raghavan, and H. Sch�tze, —"Introduction to Information Retrieval", Cambridge University Press, 2015. Link: https://nlp.stanford.edu/IR-book/				
Web Based Resources and E-books:				
https://puniversity.informaticsglobal.com/login				
Topics relevant to the development of SKILLS: Recommendation Techniques, Content-based Filtering for Skill Development through Participative Learning techniques. This is attained through assessment component mentioned in course handout.				

